



DEALER BEST PRACTICE SERIES

Failure Analysis Process – A Key  
Tool to Managing Quality

Application      Maintenance      Component Rebuild      Component Life Management      MARC Management

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## 1.0 Introduction

Nearly every Caterpillar dealership has the capability to perform failure analysis, but not many have a formalized standard process to analyze every failure that comes into the shop, identify the root causes, and communicate the findings and corrective recommendations to Caterpillar, to customers, and within their own organizations.

Root causes for most component failures come from three sources:

1. Factory defects - including parts quality, workmanship, or design weaknesses.
2. Customer actions - such as operator error, improper maintenance and monitoring, application factors, or improper removal and installation.
3. Dealer - in the form of workmanship defects or improper parts reuse.

Accurate failure analysis can:

1. Provide Caterpillar the capability to identify and correct factory problems quickly, and more effectively.
2. Cause or assist customers to modify their operation or maintenance practices, which will prevent or minimize future failures.
3. Cause or assist the CRC to modify rebuild processes to deliver higher rebuild quality.

This Best Practice is does not describe how to conduct a failure analysis. It does describe how to establish and use a formalized Failure Analysis program as a valuable tool to improve quality and customer satisfaction.

## 2.0 Best Practice Description

Several dealers have a formal failure analysis process and a standard practice to analyze every component failure that comes into the shop. The program requires a Failure Analysis staff of 3 to 6 trained, full time failure analysis specialists. These dealers prepare over 1000 failure analysis reports a year.

The key elements of an Applied Failure Analysis program include:

- **Staff.** Specialists who have completed Caterpillar AFA I, II, and III classes.
- **Laboratory.** Equipped with stereo microscope, digital camera with macro capability, micrometers, hardness tester, dye crack detection, cleaning equipment, hand tools, workbenches, good lighting, storage racks and cabinets. Ideally, the lab should be located in or near the component disassembly area.
- **Database.** A record keeping system ideally should be able to store failure analysis reports and sort records by multiple factors, such as by model, by component, by customer, by serial number, etc. It also needs to utilize standard report forms and documentation, document communications and feedback. Several dealers use a Lotus Notes system for this purpose.
- **Process.** All failures are analyzed for root cause. Personnel job descriptions and responsibilities are clearly defined. Communication procedures are established for providing analysis results and recommendations. A process to track feedback and corrective actions is established.

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Some of the benefits experienced from a standardized AFA program are significant:

- **Warranty and Policy.** Failure analysis reports are provided to warranty clerk to facilitate claim preparation and SIMS data entry. The reports improve warranty claim submission with better cost recovery.
- **Product Problem Identification and Reporting.** Reoccurring problems are summarized and reported to Caterpillar with detailed failure analysis information. Early identification leads to faster Caterpillar response in support programs, interim measures, and engineering solutions.
- **Improve Customer Fleet Availability and Operation Costs.** Identification the root causes of failures allows corrective measures to be taken to eliminate or reduce future repeat occurrences. Improvements in maintenance, machine monitoring, and operation can often result from AFA.
- **Customer Relations.** Failure Analysis that helps the customer obtain better performance results from his fleet builds confidence in dealer service support and in Caterpillar equipment. Long term, this leads to customer satisfaction and loyalty.
- **CRC Rebuild Processes.** Failure Analysis information provided to CRC Supervisors allow corrective measure to be implemented in the rebuild process, leading to higher rebuild quality and lower re-do rates.

### 3.0 Implementation Steps

1. Define staff requirements and job descriptions to include roles and responsibilities.  
AFA Technicians: Perform professional failure analysis to determine root cause.  
AFA Supervisor: Use AFA to take corrective actions to prevent failures and increase customer machine availability and reduce costs
2. Establish a Failure Analysis workstation located in or near the component disassembly area. It is important to that the Failure Analysis process does not slow down or interrupt the rebuild process. Failure Analysis lab equipment costs are approximately \$15,000.



**90x Stereo Microscope  
with photo attachment**



**Digital Camera & Micrometers**



**Storage Rack & Tool Cabinet**

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Quality**

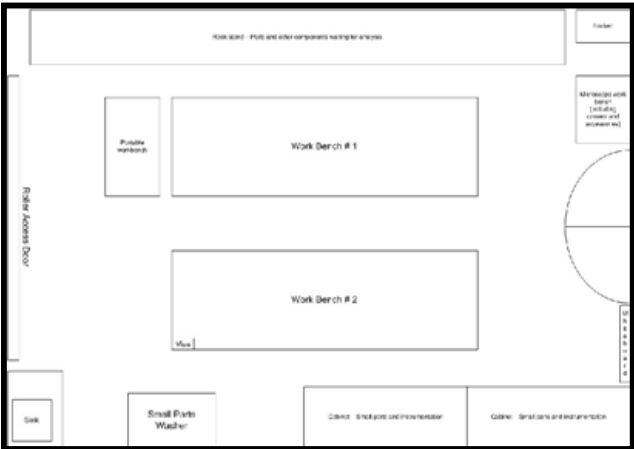
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Workbench



Lab Layout (12'x16')  
See Attachments for Details



Parts Cleaning Tank



Lighted Magnification

|                                                                                                                 |                  |              |                         |
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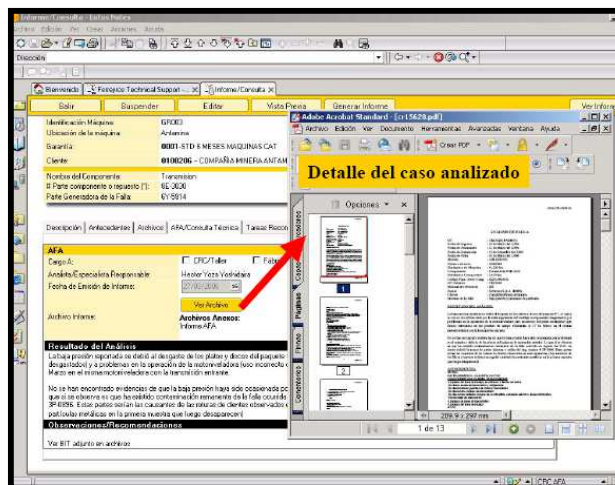
3. Develop a database able to document failure analysis reports and sort information by machine model, serial number, component, customer, and other variables. System should also track all communications and feedback. Many dealers use a Lotus Notes system for this purpose.

|                         |                                 |
|-------------------------|---------------------------------|
| WO                      | : CR13558                       |
| Reception Date          | : July 12 <sup>th</sup> 2005    |
| Installation Date       | : June 23 <sup>rd</sup> 2004    |
| Date of failure         | : July 02 <sup>nd</sup> 2005    |
| Model                   | : 793C (HT041)                  |
| Serial number           | : 4GZ00259                      |
| SMH of Machine          | : 24,272 hrs.                   |
| Component               | : Engine 3516B                  |
| SMH of Component        | : 6,968 hrs.                    |
| Fsaa Code / Comp Series | : 7TR01103 / ATY100082033       |
| Previous WO             | : CR10543                       |
| Reference Manual        | : SIS                           |
| Dealer                  | : Ferreyros S.A.A. (R080)       |
| Customer                | : Compañía Minera Antamina      |
| Symptom                 | : Loud Noise – Extremely severe |

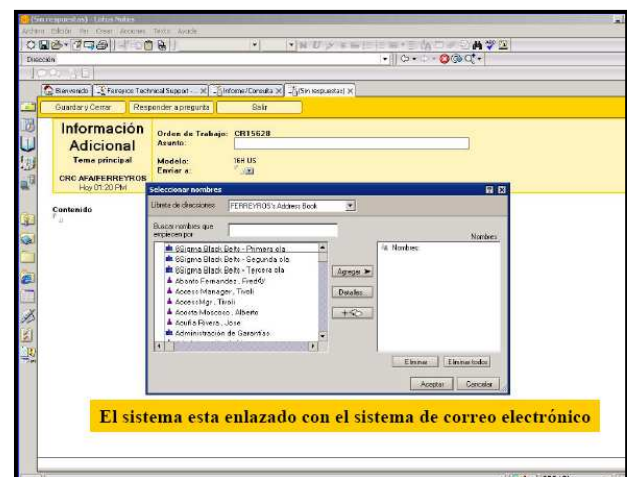
Sample of Standard Report  
Data Fields

Información de los casos puede ser  
Ordenado utilizando diferentes variables

Lotus Format  
Sort by Fields



Document, Track, Edit  
Failure reports



Communicate  
Standard Address Lists

4. Communication Process. It is important to develop a standard communication process to collect needed information for the failure analysis, share the results and recommendations, and document preventative measures taken to avoid future failures. This communications process ideally should be defined with a detailed process map. Key communications include:

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- Collecting information for analysis.
  - i. Mine records including PM history, operator reports, fluid consumption, machine inspection reports and backlog, hours on component since last rebuild and total life hours, R & I checklist, VIMS data, and operator report.
  - ii. Dealer records. Service history, delivery records, SOS and ISO information, and Caterpillar technical communication.
- Communicating results and recommendations;
  - i. To the minesite may include Maintenance management, Planning department, operations department, and admin department.
  - ii. To dealer personnel may include warranty administration, technical communicator, and CRC supervisors.
- Documenting corrective measures taken. It is important to have a feedback process that documents corrective measures taken to prevent future failures. This is vital and is the main purpose of a Failure Analysis program.

5. Standardized Failure Analysis Report. (See Attachments below for an example)

- Component identification information, serial number, hours, etc.
- Supporting data and information – service history, VIMS data, etc.
- Failed parts analysis with photos.
- Conclusions – root cause identification.
- Recommendations

#### 4.0 Benefits

- Improves warranty claim process and payment.
- Provides basis for policy or goodwill claims.
- Help Caterpillar to identify and resolve technical problems.
- Improves CRC rebuild process and quality.
- Gives customer better machine availability.
- Gives customer lower machine repair costs.
- Builds customer relations and loyalty.

#### 5.0 Resources Required

- Staff requirement (3-6 full time failure analysis specialists)
- Failure Analysis lab and equipment - \$15,000.
- Record Keeping System
- Office and furniture

#### 6.0 Supporting Attachments / References

- [Cr 13558 translation.pdf](#) - A Sample AFA Report  
[AFA Lab.jpg](#) - AFA Bay Design layout  
[AFA presentation 2007.pdf](#) - 6Sigma Project PowerPoint describing the project & process (in Spanish only)

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## 7.0 Related Best Practices

None at this time.

## 8.0 Acknowledgements

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